

```
In [1]: import pandas as pd

df = pd.read_csv("heart.csv")
print(df.head())
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang
0	52	1	0	125	212	0	1	168	0
1.0		2							
1	53	1	0	140	203	1	0	155	1
3.1		0							
2	70	1	0	145	174	0	1	125	1
2.6		0							
3	61	1	0	148	203	0	1	161	0
0.0		2							
4	62	0	0	138	294	1	1	106	0
1.9		1							

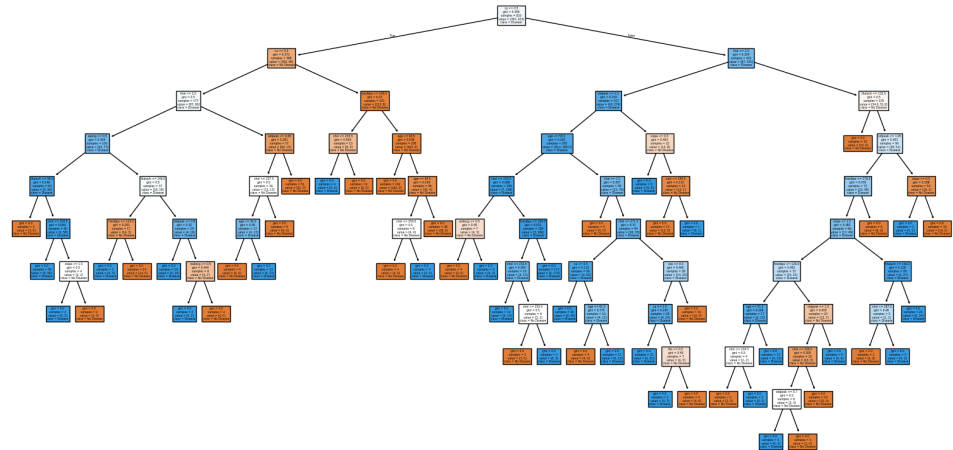
	ca	thal	target
0	2	3	0
1	0	3	0
2	0	3	0
3	1	3	0
4	3	2	0

```
In [2]: from sklearn.model_selection import train_test_split
X=df.drop("target", axis=1)
y=df['target']
X_train,X_test,y_train,y_test =
train_test_split(X,y,test_size=0.2,random_state=42)
```

```
In [3]: from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
```

```
Out[3]: ▾ DecisionTreeClassifier ⓘ ?
        ► Parameters
```

```
In [4]: #Visualize tree
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt
plt.figure(figsize=(20,10))
plot_tree(dt,feature_names=X.columns,
          class_names=["No Disease","Disease"],
          filled=True)
plt.show();
```



```
In [5]: #2.Analyze Overfitting and Control Tree Depth
from sklearn.metrics import accuracy_score
train_pred = dt.predict(X_train)
test_pred = dt.predict(X_test)

print("Train Accuracy:",
      accuracy_score(y_train, train_pred))

print("Test Accuracy:",
      accuracy_score(y_test, test_pred))
```

Train Accuracy: 1.0
Test Accuracy: 0.9853658536585366

```
In [6]: dt_depth = DecisionTreeClassifier(
          max_depth=3,
          random_state=42
        )

dt_depth.fit(X_train, y_train)
print("Train Accuracy:",
      dt_depth.score(X_train,y_train))

print("Test Accuracy:",
      dt_depth.score(X_test,y_test))
```

Train Accuracy: 0.8512195121951219
Test Accuracy: 0.7804878048780488

```
In [7]: #random forest
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
rf.fit(X_train,y_train)
rf_pred = rf.predict(X_test)

print(
    "Random Forest Accuracy:",
    accuracy_score(y_test, rf_pred)
)
```

Random Forest Accuracy: 0.9853658536585366

```
In [9]: #4. Interpret Feature Importance
import pandas as pd

importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": rf.feature_importances_
})

importance = importance.sort_values(
    by="Importance",
    ascending=False
)

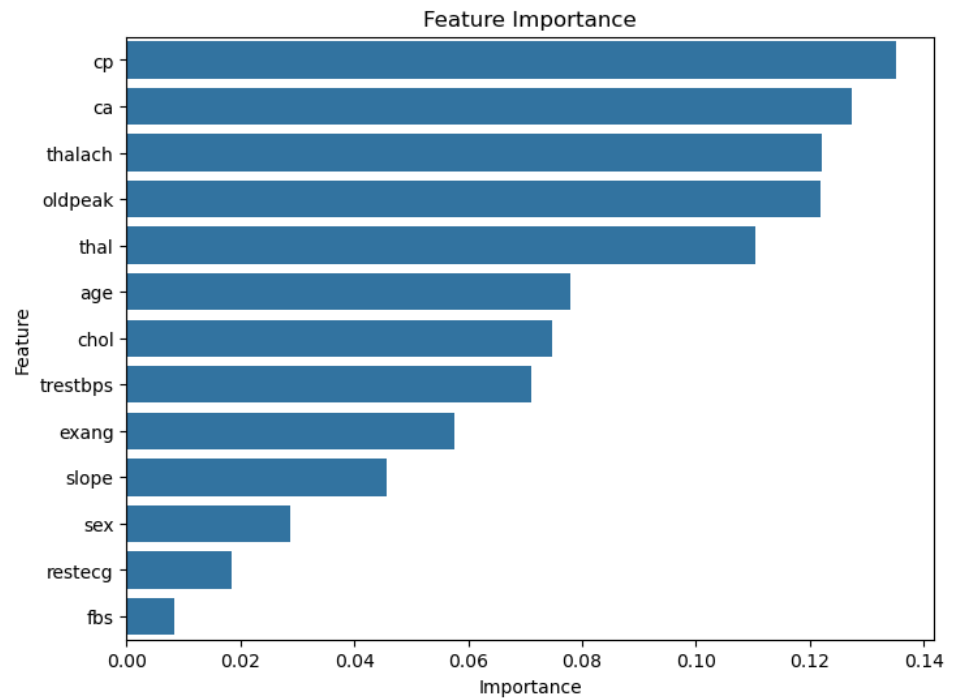
print(importance)
import seaborn as sns

plt.figure(figsize=(8,6))

sns.barplot(
    x="Importance",
    y="Feature",
    data=importance
)

plt.title("Feature Importance")
plt.show()
```

	Feature	Importance
2	cp	0.135072
11	ca	0.127327
7	thalach	0.122169
9	oldpeak	0.121905
12	thal	0.110518
0	age	0.077908
4	chol	0.074822
3	trestbps	0.071171
8	exang	0.057594
10	slope	0.045782
1	sex	0.028731
6	restecg	0.018557
5	fbs	0.008444



```
In [10]: #5. Evaluate Using Cross Validation
from sklearn.model_selection import cross_val_score
scores = cross_val_score(
    rf,
    X,
    y,
    cv=5
)
print(scores)
print(
    "Mean CV Accuracy:",
    scores.mean()
)
```

```
[1. 1. 1. 1. 0.98536585]
Mean CV Accuracy: 0.9970731707317073
```